Population: normal with $\mu = 4.11$, $\sigma = 0.18$

① $P(X \leq 4.01)$

② $P(3.73 \leq X \leq 4.28)$

Step 1: write down the probabilities as $P(X \leq 4.01)$

Step 2: calculate z-value. $z = \frac{x - \mu}{\sigma} = \frac{4.01 - 4.11}{0.18}$

$= -0.56$

Step 3: write down an equation

$P(X \leq 4.01) = P(Z \leq -0.56)$

Step 4: Check.

① check if $z > 0$:

if $z > 0$: ✓,

if $z < 0$: use symmetry:

$P(Z \leq -0.56) = P(Z \geq 0.56)$

② check if "≤".

↳ if "≤": ✓

↳ if "≥": use $P(Z \geq a) = 1 - P(Z < a)$.

↳ now $P(Z \geq 0.56) = 1 - P(Z < 0.56)$

Step 5:

$$P(X \leq 4.01) = P(Z \leq -0.56) = P(Z \geq 0.56) = 1 - P(Z < 0.56)$$

I

① either "≤" or "<".

② $0.56 > 0$.

check the normal table: $P(Z < 0.56) = 0.7123$.

$\therefore P(X \leq 4.01) = 1 - 0.7123 = 0.2877$.

$P(3.73 \leq X \leq 4.24) = P(X \leq 4.24) - P(X \leq 3.73)$

PSTAT 5A | **Discussion 6 Review 1** **Fall 2026**

Problems 1–3: Sampling distributions and confidence intervals

Work with a partner. Fill in the key ideas first, then use them to solve D6 Problems 1–3. Focus on choosing the correct method and explaining why.

1. Sampling Distribution of the Sample Mean (D6 Problem 1)

Suppose a random sample of size n is drawn from a population with mean μ and standard deviation σ.

1. Complete: $E(\bar{X}) =$ _____ $SD(\bar{X}) =$ _____.
2. When is $\bar{X}$ normally distributed or approximately normal? Complete both conditions.
   - The population distribution of X is _____.
   - The population distribution is unknown, but $n \geq$ _____ (Central Limit Theorem).
3. Complete the sampling distribution, using **standard deviation** as the second parameter:

$$\bar{X} \overset{\text{approx}}{\sim} N(\text{_____}, \text{_____}) \quad \text{(when CLT applies).}$$

4. To find $P(\bar{X} > a)$ or $P(a < \bar{X} < b)$, standardize:

$$Z = \frac{\bar{X} - \text{_____}}{\text{_____}}.$$

Discuss: If the sample size increases, does the spread of $\bar{X}$ increase or decrease? Why?

2. Confidence Interval for a Population Mean (D6 Problems 2–3)

For a random sample from a normal population, a confidence interval has the form | point estimate $\pm$ critical value $\times$ standard error |

	Population SD σ known	Population SD σ unknown
Distribution / critical value	_____	_____
Standard error	_____	_____
Confidence interval for μ	$\bar{x} \pm$ _____	$\bar{x} \pm$ _____

1. If σ is unknown and $n = 25$, the degrees of freedom are $df =$ _____.
2. For a 98% confidence interval, $\alpha =$ _____ and each tail has area _____.
3. Before choosing a method, what two facts should you check about the sample and the population?

_____

3. Think, Compare, Explain (D6 Problems 2–3)

1. With the same $\bar{x}$, n, and numerical SD, which 98% interval is wider: t or Z? _____ Why?

_____

2. If the confidence level increases from 98% to 99%, the critical value _____ and the interval becomes _____.

Today: D6 Problems 1–3. Review 2 will cover the remaining problems on Wednesday.

$\bar{X} \pm z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$ or $\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}}$

Case: population sd: σ is known

z: normal dist. → check normal table

Case: σ is unknown.

sample var: s^2.

t_{n-1}: t dist.

$n-1$: degree of freedom

check t table.

1. A random sample of 48 observations is drawn from a population with a mean of 512 and a standard deviation of 32.5.

a) State the distribution of the sample mean, $\bar{X}$. (Show any conditions that you have checked.)

$E(\bar{X}) = 512.$

$sd(\bar{X}) = \frac{32.5}{\sqrt{48}}$

$n \geq 30 \rightarrow n = 48 > 30$

CLT. $\bar{X} \overset{\text{approx}}{\sim} N\left(512, \frac{32.5}{\sqrt{48}}\right)$

b) Calculate the probability that the sample mean is more than 510.

$P(\bar{X} > 510) = P\left(Z > \frac{510 - 512}{\frac{32.5}{\sqrt{48}}}\right) = P(Z > -0.43)$

$= P(Z < 0.43) = 0.6664$

c) Calculate the probability that the sample mean is between 515 and 517.

$P(515 \leq \bar{X} \leq 517)$

$= P\left(\frac{515 - 512}{\frac{32.5}{\sqrt{48}}} \leq Z \leq \frac{517 - 512}{\frac{32.5}{\sqrt{48}}}\right)$

$= P\left(Z \leq \frac{517 - 512}{\frac{32.5}{\sqrt{48}}}\right) - P\left(Z < \frac{515 - 512}{\frac{32.5}{\sqrt{48}}}\right)$

$= P(Z \leq 1.07) - P(Z < 0.64)$

$= 0.8577 - 0.7389$

$= 0.1188.$

CI for population mean where population mean is unknown.

$n=25$

2. A random sample of 25 observations is drawn from a population with a normal distribution. If the sample mean is 88.1 and the sample variance is 54.76, find the 98% confidence interval for the population mean.

$\bar{x}=88.1$ $S^2=54.76$ 98% CI for μ?

the population mean is unknown.

so we use a t-distribution with $df = n-1 = 24$.

$\frac{\alpha}{2} = \frac{1-0.98}{2} = 0.01$. check the table: $t_{0.01,24} = 2.492$.

so the CI is $\bar{x} \pm t_{\frac{\alpha}{2},n-1} \times \frac{s}{\sqrt{n}} = 88.1 \pm 2.492 \cdot \left(\frac{\sqrt{54.76}}{\sqrt{25}}\right)$

$= (84.4118, 91.7882)$

$n=25$ population variance $\sigma^2=54.76$ $\bar{x}=88.1$

3. A random sample of 25 observations is drawn from a population with a normal distribution with a variance of 54.76. If the sample mean is 88.1, find the 98% confidence interval for the population mean. Compare this confidence interval with the one found in question 2. Next, find the 99% confidence interval for the population mean. How does this compare to the 98% confidence interval?

we should use Z-interval since population variance is known.

$\frac{\alpha}{2} = \frac{1-0.98}{2} = 0.01$. check table: $z_{0.01} = 2.326$.

$\bar{x} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} = 88.1 \pm 2.326 \cdot \left(\frac{\sqrt{54.76}}{\sqrt{25}}\right)$

$= (84.6575, 91.5425)$.

Compare to Problem 2:

98% Z-interval is narrower than 98% t-interval.

because estimating population standard deviation introduces extra

uncertainty, making the t-CI wider than z-CI.

99% CI: $\frac{\alpha}{2} = \frac{1-0.99}{2} = 0.005$. $z_{0.005} = 2.576$.

$\therefore$ 99% CI, $\bar{x} \pm z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{n}} = 88.1 \pm 2.576 \times \frac{\sqrt{54.76}}{\sqrt{25}}$

$= (84.2875, 91.9125)$.

99% CI is wider than 95% CI because a higher CI requires a larger central probability under the curve. ——.